

BUKHARA INTL, Uzbekistan

WMO: 381415

Lat: 39.775N

Lon: 64.483E

Elev: 751

StdP: 14.30

Time Zone: 5.00 (E05)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.8	15.9	1.0	5.9	13.1	6.9	8.1	19.5	20.8	30.0	18.0	38.7	6.4	340	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.6	104.4	67.3	102.2	66.2	100.1	65.3	69.9	98.7	68.3	96.4	67.1	94.5	11.3	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.4	77.8	77.2	57.6	72.9	76.7	56.8	70.6	76.5	34.2	98.0	32.8	97.3	31.8	94.9	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	17.8	16.0	8.2	109.4	7.3	2.8	3.0	111.4	-1.2	113.1	-5.3	114.6	-10.6	116.7
			7.5	72.9	7.3	1.9	2.3	74.3	-2.0	75.4	-6.1	76.4	-11.4	77.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	61.1	35.8	39.3	52.5	63.1	74.8	82.8	85.5	81.7	71.9	59.1	47.0	38.0	(6)
(7)		DBStd	19.21	10.19	10.61	8.73	8.32	5.18	4.69	4.39	5.11	5.81	7.86	9.23	8.31	(7)
(8)		HDD50	1401	448	316	75	7	0	0	0	0	0	18	157	380	(8)
(9)		HDD65	3756	906	721	394	130	2	0	0	0	10	211	545	837	(9)
(10)		CDD50	5445	7	16	154	401	768	984	1102	981	657	301	66	8	(10)
(11)		CDD65	2325	0	0	8	74	305	534	637	517	217	29	4	0	(11)
(12)		CDH74	32458	0	2	145	998	3891	7776	9410	7000	2701	486	48	1	(12)
(13)		CDH80	18720	0	0	35	374	1937	4805	5994	4179	1245	138	13	0	(13)
(14)	Wind	WSAvg	7.8	6.7	7.6	7.8	7.7	7.9	8.8	10.0	9.3	8.3	6.4	5.9	6.5	(14)
(15)	Precipitation	PrecAvg	5.7	0.7	0.9	1.1	0.9	0.4	0.1	0.0	0.0	0.1	0.6	0.7	(15)	
(16)		PrecMax	8.1	1.9	2.5	1.9	2.4	2.1	0.4	0.2	0.1	0.2	0.6	2.5	1.5	(16)
(17)		PrecMin	2.9	0.2	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	(17)
(18)		PrecStd	1.6	0.4	0.6	0.5	0.6	0.5	0.1	0.0	0.1	0.2	0.6	0.6	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.0	70.1	84.3	95.1	100.1	106.2	109.4	105.9	98.6	89.4	80.8	66.6	(19)
(20)		MCWB	48.1	53.5	57.9	62.4	63.4	66.2	69.3	68.2	64.6	59.9	57.9	48.1	(20)	
(21)		2%	DB	59.2	63.2	77.4	88.2	96.7	103.7	105.9	102.2	94.7	84.0	71.5	59.3	(21)
(22)		MCWB	46.4	50.4	56.4	60.6	62.8	65.9	68.2	66.7	62.6	57.6	53.3	46.6	(22)	
(23)		5%	DB	53.9	57.6	73.1	84.2	93.5	100.6	103.6	100.0	91.3	79.1	66.1	55.0	(23)
(24)		MCWB	44.9	47.4	54.4	59.0	61.9	64.6	67.4	65.9	61.7	56.3	51.6	45.5	(24)	
(25)		10%	DB	49.6	53.8	68.0	80.2	90.0	98.4	100.4	96.9	87.6	75.2	61.2	50.0	(25)
(26)		MCWB	42.9	46.1	53.1	57.8	61.5	64.0	65.9	64.5	60.4	55.5	49.2	43.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.2	54.8	60.8	65.6	67.1	69.4	72.6	72.2	66.4	62.0	58.5	50.6	(27)
(28)		MCDWB	59.8	66.2	74.1	82.5	88.6	97.2	102.5	98.6	92.0	83.0	74.6	62.1	(28)	
(29)		2%	WB	47.9	51.8	58.3	63.0	65.6	67.9	70.5	69.3	64.4	59.6	55.5	47.9	(29)
(30)		MCDWB	58.0	60.4	72.8	80.4	86.7	95.9	99.8	96.8	89.8	78.9	68.3	56.7	(30)	
(31)		5%	WB	45.5	49.2	56.0	61.5	64.4	66.6	69.0	67.6	62.8	57.8	52.7	45.7	(31)
(32)		MCDWB	52.8	56.7	69.3	78.1	85.5	94.4	98.3	94.6	87.6	75.6	63.2	52.8	(32)	
(33)		10%	WB	42.7	46.5	54.0	59.8	63.2	65.5	67.6	66.1	61.3	56.1	50.2	43.5	(33)
(34)		MCDWB	48.8	52.3	66.6	75.3	84.2	92.8	96.3	92.9	85.1	72.9	60.0	49.6	(34)	
(35)	Mean Daily Temperature Range	MDBR	16.4	17.4	21.2	23.6	26.5	27.7	27.6	28.0	28.2	26.3	21.0	16.3	(35)	
(36)		5% DB	MCDBR	24.3	24.4	29.5	30.7	31.4	31.3	30.9	30.7	32.3	31.7	28.0	24.2	(36)
(37)		MCWBR	14.3	13.9	12.1	10.4	9.8	9.9	11.7	12.1	12.5	13.4	13.8	14.1	(37)	
(38)		5% WB	MCDBR	22.0	20.9	25.8	25.1	26.5	28.2	29.0	28.8	30.2	27.9	24.0	21.5	(38)
(39)		MCWBR	13.7	12.7	12.0	9.5	9.1	9.3	11.2	11.8	12.2	12.5	12.5	13.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.381	0.421	0.491	0.521	0.516	0.509	0.474	0.443	0.423	0.464	0.440	0.383	(40)	
(41)		taud	2.122	2.021	1.876	1.840	1.874	1.895	1.992	2.061	2.071	1.930	1.979	2.100	(41)	
(42)		Ebn at Noon	237	244	239	242	246	248	256	261	257	225	210	223	(42)	
(43)		Edh at Noon	35	45	58	64	63	62	56	50	47	49	41	34	(43)	
(44)	All-Sky Solar Radiation	RadAvg	625	927	1311	1770	2177	2465	2438	2228	1818	1256	790	556	(44)	
(45)		RadStd	55	92	103	143	95	73	53	46	47	67	74	62	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (6 neighbors)	N/A	N/A	N/A	+0.58	N/A	-1.44	N/A	N/A	N/A	+108	(47)

Nomenclature: See separate page